Event-Local Sheafification and a Linear/Non-Linear Adjunction with Chu Coefficients

Hoapa project

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Extended abstract — machine-checked in cubical Agda

Abstract

We construct an event-local sheafification functor and use it to lift a Benton linear/non-linear adjunction over a small site of finite causal interaction contexts. The objects of the site are finite local views of a concurrent interaction: unityped actors, the events in which they participate, and the causal order on those events. The topology is generated by decomposing each context into its single-event subcontexts. For any coefficient category with finite products, sheafification sends a presheaf to the product of its values on those subcontexts and is left adjoint to the inclusion of event-local sheaves. Given a Benton adjunction $F \dashv G$ between a cartesian coefficient category and a $*$-autonomous coefficient category, this construction yields an adjunction between cartesian-valued and linear-valued event-local sheaves. The cartesian side is the freely copyable-and-discardable regime, the $*$-autonomous side is the resource-sensitive regime with linear duals, and the induced comonad is the sheaf-level linear exponential "!". The hypotheses are also jointly realized in a concrete coefficient model: $\mathsf{V} = \mathrm{Chu}(\mathbf{Set}, \mathsf{Bool})$ has the required finite products and a minimal $*$-autonomous interface, and the Benton adjunction $\mathbf{Set} \rightleftarrows \mathrm{Chu}(\mathbf{Set}, \mathsf{Bool})$ is constructed in Agda. Separately, the postulate-free session syntax is interpreted into this Chu model by an underlying functor. Thus the sheaf-level theorem is generic, but not vacuous.

This note isolates one theorem from a larger programme that models concurrent interaction systems, and the typed data produced by them, in the language of sheaves and adjoint functors. The theorem starts from a deliberately austere view of agency. Actors are unityped: mathematically, they are elements of a bare set, and the formal system exposes no role, origin, or hidden state on which a construction can branch. The structure lies instead in events, causal dependence, incidence between events and actors, and the typed values assigned to finite contexts of interaction. The result below says that a coherent global assignment over such a context is freely reconstructed from its event-local assignments. The system's global point of view is not imposed from outside; it is the coherent reconstruction of its event-local standpoints.

The theorem has two parts. First, event-local observations freely generate a coherent global object. Second, that reconstruction respects the cartesian-linear structure of the values assigned to contexts: reusable data live on the cartesian side, resource-sensitive session-like data live on the linear side, and the Benton adjunction relates the two at the sheaf level. Thus actor-uniformity belongs to the site, while resource-sensitivity belongs to the coefficient category; the construction keeps the two as independent axes.

The event-local site

The actor model of concurrency [1] describes computation as a collection of agents that interact solely by exchanging messages. We use only the unityped aspect of that tradition: actors are elements of a bare set, and the kernel exposes no role, origin, or tag on which a construction can branch. Lamport's causal order [2] supplies the temporal structure: events are partially ordered by possible influence. Following presheaf models of concurrency [3], these finite observations are organized as a small site.

Fix a set A of actors. A *finite causal context* over A consists of a finite set of events, a partial order on those events, and an incidence relation recording which actors participate in which events. The context is *supported*: every actor present in the context participates in at least one event. A morphism of contexts is a causality-reflecting embedding, preserving actors, events, and incidence, and both preserving and reflecting the causal order. Write Ctx_A for the resulting category; we call it the actor-causal site.

The topology used here is event-local. For an event e of a context F, write F_e for the single-event subcontext at e, whose actors are exactly those incident to e. The generating cover of F is

$$\{\, F_e \longrightarrow F \,\}_{e \in E(F)}.$$

No other covers are made primitive in this theorem. In particular, causal refinement, higher overlaps, and richer compatibility conditions are deferred to later topologies. The Agda development inherited the internal word "point" for these single-event subcontexts; in this abstract we use *event-local* for the site-side notion and reserve "points" and "states" for Chu spaces below.

Event-local sheafification

Let $\mathcal{C}$ be a coefficient category with finite products. A $\mathcal{C}$-valued presheaf on the site is a functor

$$P : \mathsf{Ctx}_A^{\mathrm{op}} \longrightarrow \mathcal{C},$$

and we write $\mathsf{PSh}_A(\mathcal{C}) = [\mathsf{Ctx}_A^{\mathrm{op}}, \mathcal{C}]$. An event-local sheaf is a presheaf satisfying descent for the single-event covers above. The formalization proves the expected product-locality characterization: a presheaf is event-local precisely when its value on a context is determined by its values on the single-event subcontexts of that context.

The inclusion $\iota_{\mathcal{C}} : \mathsf{Sh}_A^{\mathrm{ev}}(\mathcal{C}) \hookrightarrow \mathsf{PSh}_A(\mathcal{C})$ has an explicit left adjoint. For a presheaf P, define

$$a_{\mathcal{C}}(P)(F) = \prod_{e \in E(F)} P(F_e),$$

with restrictions induced by the corresponding maps between single-event subcontexts. This is sheafification in the restricted sense used here: the reflector onto presheaves satisfying descent for the explicitly presented event-local covers. It is not claimed to be the associated-sheaf functor for an independently formalized Grothendieck coverage. The machine-checked result is the adjunction

$$a_{\mathcal{C}} \dashv \iota_{\mathcal{C}}.$$

The linear/non-linear lift

Let C be a cartesian category and let V be a $*$-autonomous category [4], both with finite products. A linear/non-linear adjunction in the sense of Benton [5] is a monoidal adjunction $F \dashv G : \mathsf{C} \rightleftarrows \mathsf{V}$ from the cartesian world to the linear one. The induced comonad on V is the linear exponential "!". The Agda interface used by the theorem is intentionally minimal: it records the two functors, the adjunction, and the strong-monoidal comparison maps of F. The proof consumes the adjunction and product structure; the omitted global coherence and tensor-hom universal property are not unfolded.

Postcomposition lifts F and G to presheaves,

$$F_* : \mathsf{PSh}_A(\mathsf{C}) \to \mathsf{PSh}_A(\mathsf{V}), \qquad G_* : \mathsf{PSh}_A(\mathsf{V}) \to \mathsf{PSh}_A(\mathsf{C}),$$

with $F_* \dashv G_*$ computed pointwise. The right adjoint preserves event-local sheaves: descent for the single-event cover is transported across the hom-set bijection $\mathsf{C}(X, GY) \cong \mathsf{V}(FX, Y)$. The left adjoint is made to land in sheaves by composing with the reflector. With

$$L = a_{\mathsf{V}} \circ F_* \circ \iota_{\mathsf{C}}, \qquad R = G_*|_{\mathsf{Sh}^{\mathrm{ev}}_A(\mathsf{V})},$$

we obtain the theorem.

Theorem. *Let C be a cartesian category and V a $*$-autonomous category, both with finite products, and let $F \dashv G : \mathsf{C} \rightleftarrows \mathsf{V}$ be a Benton linear/non-linear adjunction. Then there is an adjunction*

$$L \dashv R : \mathsf{Sh}^{\mathrm{ev}}_A(\mathsf{C}) \rightleftarrows \mathsf{Sh}^{\mathrm{ev}}_A(\mathsf{V})$$

between cartesian-valued and linear-valued event-local sheaves on the actor-causal site. The induced comonad $L \circ R$ is the sheaf-level linear exponential.

A concrete Chu coefficient model

The theorem above is generic in its coefficient categories. The new Chu construction changes its status: the hypotheses are not merely assumed, but are simultaneously inhabited in a concrete machine-checked model. For a dualizing set k, a Chu object over **Set** is a triple $(A^+, A^-, \langle -, - \rangle : A^+ \times A^- \to k)$ [6, 7]. In the standard Chu terminology, A^+ is the carrier of points and A^- the carrier of states; this is distinct from the event-local subcontexts of the site.

At $k = \mathsf{Bool}$, the Agda development constructs the category $\mathrm{Chu}(\mathbf{Set}, \mathsf{Bool})$, its finite products, and the formalization's minimal $*$-autonomous interface. It also constructs the Benton adjunction

$$\mathbf{Set} \rightleftarrows \mathrm{Chu}(\mathbf{Set}, \mathsf{Bool}),$$

with right adjoint $G(A) = A^+$ and left adjoint $F(X) = (X, \mathsf{Bool}^X, \mathrm{ev})$. Supplying these data to the generic event-local theorem yields a concrete witness of the lifted adjunction $L \dashv R$. In this construction, the Benton linear/non-linear adjunction is the cartesian-linear bridge: it relates reusable cartesian values to resource-sensitive linear values. The Chu model supplies concrete coefficients in which that bridge exists, so the sheaf-level adjunction is an instantiated theorem rather than a purely conditional one.

Session syntax in the model

A separate result connects the session syntax to this coefficient model. The postulate-free syntax of session types and their structural morphisms, in the linear-propositions tradition for session types [8, 9], is interpreted into any target with the formalization's minimal $*$-autonomous and finite-product interface by an object map $J0$, a duality theorem $J0(S^{\perp_s}) \cong (J0(S))^{\perp}$, and a morphism map $J1$. Instantiating that interpretation at Chu($\mathbf{Set}, \mathsf{Bool}$) gives an underlying functor in the Cubical Agda category library from the session category to the Chu coefficient category. This is intentionally weaker than full session semantics: it is not an operational send/receive model, not a coherent monoidal or $*$-autonomous session functor, and not a construction of the relevant cyclic operadic action in Chu. The remaining action postulate in `Sessions.agda` is therefore not discharged by this result.

Scope and significance

Three classical lines meet in the theorem. It is a descent statement over a small site, in the tradition of Grothendieck [10, 11]. It is an instance of Lawvere's principle that adjointness organizes foundational structure [12]: the global object is obtained by a left adjoint, and its universal property is the freeness of the reconstruction. And it lifts Benton's linear/non-linear adjunction from coefficient categories to sheaves. The Chu instance adds non-vacuity: the general theorem is still stated parametrically, but there is now a concrete postulate-free coefficient model in which its hypotheses hold.

In the wider programme, this site/coefficient separation fixes a bounded piece of the operational side of a larger conjectural architecture. Event-local observations freely generate a coherent global object, that reconstruction is compatible with resource-sensitive values, and one concrete Chu coefficient model realizes the required value-side structure. Claims about richer causal descent, operational session semantics, formal introduction of external data into the system, and cross-site modal adjunctions belong to later stages.

Related work

The theorem has siblings by facet rather than a single direct predecessor. The local-to-global pattern is close to Abramsky and Brandenburger's sheaf-theoretic account of contextuality [13], where contextuality is detected as an obstruction to global sections, and to later cohomological developments of that line [14]. Event-based sheaf models of systems appear in work of Zardini, Spivak, Censi, and Frazzoli [15]. Local views of distributed computation are organized topologically in work of Goubault, Ledent, and Rajsbaum [16].

On the linear-concurrency side, thin concurrent games and generalized species provide a recent adjacent model [17], while work on session types derived from linear logic clarifies the session-theoretic side [9, 18]. Licata, Shulman, and Riley's fibrational framework for substructural and modal logics [19], and the modal homotopy type theory of Riley, Finster, and Licata [20], are further siblings on the type-theoretic side: they organize movement between logical regimes rather than collapsing those regimes into one. These works share pieces of the structure below, but do not construct the actor-causal site, event-local sheafification, Benton lift, and Chu coefficient instance proved here.

Navigating the Agda formalization

The repository is:

`https://codeberg.org/doc2cog/hoapa`

The theorem is checked in the subtree:

`formal/Concurrency/`

The most direct way into the proof is to start from the theorem module:

`PointLinearNonlinearAdjunction.agda`

then follow the imports backwards through the reflector, the sheaf adapter, the site geometry, and finally the coefficient models.

The formalization splits into three compact layers. The role of Agda here is not only to record the statement: it checks that the reflector, lifted adjunction, finite-product witnesses, and Chu instance inhabit the same interfaces simultaneously.

Because the development is cubical, the function extensionality needed for naturality, functoriality, and the adjunction triangle identities is available as a theorem rather than as an extra postulate.

Event-local reflector and lifted adjunction. Folder: `formal/Concurrency/Hybrid/`

`SupportedChangeOfBase.agda` `PointSheafAdjunction.agda`

`SupportedLinearNonlinearSheafAdapter.agda` `PointLinearNonlinearAdjunction.agda`

Event-local site geometry. Folder: `formal/Concurrency/Sites/`

`EventPoint.agda` `EventSupported.agda` `PointCover.agda`

`PointOverlap.agda` `SupportedFiniteFrameCategory.agda`

Concrete Chu coefficient stack. Folders:

- `formal/Concurrency/Algebras/`
- `formal/Concurrency/Hybrid/`

`Chu.agda` `ChuFiniteProducts.agda` `ChuMonoidal.agda`

`ChuLinearNonlinear.agda` `ChuPointVProd.agda` `ChuPointExample.agda`

The final Agda witness is named `Chu-nonvacuity`; its fully expanded type is omitted because writing it out makes the type-checker expand definitions prohibitively.

Session syntax and Chu interpretation. Folder: `formal/Concurrency/Algebras/`

`SessionSyntax.agda` `SessionMorphismSyntax.agda`

`SessionObjectModel.agda` `SessionChuModel.agda`

A note on authorship

This abstract was drafted with the assistance of a language model and reviewed by the authors; the surrounding framework and its philosophical motivation are presented separately in the project repository. The formal claims reported here are backed by proofs that the cubical Agda type-checker accepts in the modules named above, with the scope limitations stated explicitly.

Limitations

We are explicit about what is not done here.

- The value-level adjunction $F \dashv G$ is assumed in the general theorem; it is discharged only in the concrete Chu instance $(\mathbf{Set}, \mathrm{Chu}(\mathbf{Set}, \mathsf{Bool}))$.

- The Chu model uses the formalization's minimal $*$-autonomous interface. The global monoidal coherence and tensor-hom universal property of Chu categories are not fully formalized here.

- The topology only imposes descent along single-event covers. Causal order, overlaps of events, and richer compatibility conditions do not yet carry descent content.

- Event-local sheaves are defined by descent for the explicitly presented single-event covers. We do not yet compare this with a fully formalized Grothendieck coverage or library sheaf condition for the same site.

- The Sessions-to-Chu interpretation is an underlying functorial realization of the syntax, not operational send/receive semantics and not a coherent monoidal or $*$-autonomous session semantics.

- This is not a cross-site modal adjunction, not a construction of the required cyclic operadic action in Chu, and not an account of the formal introduction of external data into the system.

Outlook: the arc from here

The theorem constitutes one fragment of a larger conjectural architecture, and the limitations above double as its roadmap. Four moves lead outward:

1. Richer causal descent: replace the single-event covers by a genuine Grothendieck coverage recording causal order and event overlaps.

2. Operational session semantics: upgrade the Sessions-to-Chu functor to a coherent monoidal / $*$-autonomous interpretation with operational content, discharging the remaining cyclic operadic action.

3. Introduction of external data: a formal account of how typed data enters the system from

outside, to be tested at scale in a real-life implementation built in parallel with the formal side of the framework.

4. The modal companion and the cross-site adjunction: construct the second cohesive site, of perspectives or modal structural contexts, in the cohesive homotopy-type-theoretic tradition, and the geometric morphism relating it to the actor-causal site inside one ambient ∞-topos.

That last step is the target. The two sites are conjectured to sit in a single ambient ∞-topos as an adjoint pair

$$\mathsf{Allow} \dashv \mathsf{Require} : \mathcal{M} \rightleftarrows \mathcal{K},$$

the left adjoint generative, the right reflective, whose unit $\eta : \mathrm{id} \Rightarrow \mathsf{Require} \circ \mathsf{Allow}$, iterated, is the developing network of a mathematical practice. The present theorem constructs, on the community side $\mathcal{K}$, the first event-local endpoint of that adjunction.

If homotopy type theory is the right foundation for mathematics, then its modal, cohesive extension $\mathcal{M}$ is the right home for mathematics as a practice: the development of mathematics is the free iteration of a single adjunction $\mathsf{Allow} \dashv \mathsf{Require}$ between a cohesive site of perspectives and a community of communicating agents, inside one ambient ∞-topos.

References

[1] C. Hewitt, P. Bishop, and R. Steiger. A universal modular ACTOR formalism for artificial intelligence. In *Proceedings of the 3rd International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 235–245, 1973.

[2] L. Lamport. Time, clocks, and the ordering of events in a distributed system. *Communications of the ACM* 21(7):558–565, 1978.

[3] G. L. Cattani and G. Winskel. Presheaf models for concurrency. BRICS Report Series RS-96-35, 1996. Also in *Computer Science Logic (CSL '96)*, LNCS 1258, Springer, 1997.

[4] M. Barr. $*$-*Autonomous Categories*. Lecture Notes in Mathematics 752. Springer, 1979.

[5] P. N. Benton. A mixed linear and non-linear logic: proofs, terms and models. In *Computer Science Logic (CSL '94)*, LNCS 933, pp. 121–135. Springer, 1995.

[6] M. Barr. The Chu construction. *Theory and Applications of Categories* 2(2):17–35, 1996.

[7] V. R. Pratt. The Stone gamut: a coordinatization of mathematics. In *Proceedings of the 10th Annual IEEE Symposium on Logic in Computer Science (LICS)*, pp. 444–454. IEEE, 1995.

[8] L. Caires and F. Pfenning. Session types as intuitionistic linear propositions. In *CONCUR 2010*, LNCS 6269, pp. 222–236. Springer, 2010.

[9] P. Wadler. Propositions as sessions. *Journal of Functional Programming* 24(2–3):384–418, 2014.

[10] M. Artin, A. Grothendieck, and J.-L. Verdier, eds. *Théorie des topos et cohomologie étale des schémas (SGA 4)*. Lecture Notes in Mathematics 269, 270, 305. Springer, 1972–1973.

[11] S. Mac Lane and I. Moerdijk. *Sheaves in Geometry and Logic: A First Introduction to Topos Theory*. Springer, 1992.

[12] F. W. Lawvere. Adjointness in foundations. *Dialectica* 23(3–4):281–296, 1969.

[13] S. Abramsky and A. Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics* 13:113036, 2011.

[14] S. Abramsky, R. S. Barbosa, K. Kishida, R. Lal, and S. Mansfield. Contextuality, cohomology and paradox. In *24th EACSL Annual Conference on Computer Science Logic (CSL 2015)*, LIPIcs 41, pp. 211–228. Schloss Dagstuhl, 2015.

[15] G. Zardini, D. I. Spivak, A. Censi, and E. Frazzoli. A compositional sheaf-theoretic framework for event-based systems. arXiv:2101.10485, 2021.

[16] E. Goubault, J. Ledent, and S. Rajsbaum. A simplicial complex model for dynamic epistemic logic to study distributed task computability. arXiv:1809.03095, 2018.

[17] P. Clairambault, G. Olimpieri, and F. Paquet. From thin concurrent games to generalized species of structures. arXiv:2302.01082, 2023.

[18] B. van den Heuvel and J. A. Pérez. Comparing session type systems derived from linear logic. arXiv:2401.14763, 2024.

[19] D. R. Licata, M. Shulman, and M. Riley. A fibrational framework for substructural and modal logics. In *2nd International Conference on Formal Structures for Computation and Deduction (FSCD 2017)*, LIPIcs 84, pp. 25:1–25:22. Schloss Dagstuhl, 2017. doi:10.4230/LIPIcs.FSCD.2017.25.

[20] M. Riley, E. Finster, and D. R. Licata. Synthetic spectra via a monadic and comonadic modality. arXiv:2102.04099, 2021.